

MATH 4162/Math 6201
Numerical methods for differential equations
Fall 2025

Homework problem set #1

Show all necessary steps of working in the given space

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Upload the assignment solution no later than

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Full marks 50

1. (based on week 1)[15 pts]

(a) Consider the KdV equation on $\mathbb{R}$

$$\frac{\partial u}{\partial t} - 6u \frac{\partial u}{\partial x} + \frac{\partial^3 u}{\partial x^3} = 0$$

with decay conditions $u, u_x, u_{xx} \rightarrow 0$ as $|x| \rightarrow \infty$ for each fixed t . Show that the “mass”

$$M(t) = \int_{-\infty}^{\infty} u(x, t), dx$$

is constant in time.

(b) Use the Fourier transform pair given as

$$\hat{u}(k, t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} u(x, t) e^{ikx} dx \quad \text{and} \quad u(x, t) = \int_{-\infty}^{\infty} \hat{u}(k, t) e^{-ikx} dk.$$

and consider

$$\frac{\partial u}{\partial t} + \frac{\partial u}{\partial x} = \frac{\partial^2 u}{\partial x^2} + \frac{\partial^3 u}{\partial x^3}, \quad x \in \mathbb{R}, \quad t > 0,$$

with $u(x, 0) = f(x)$ and $u, \partial_x^m u \rightarrow 0$ as $|x| \rightarrow \infty$.

Convert the PDE into a system of ODE and finally, using inverse Fourier transform, show that

$$u(x, t) = \int_{-\infty}^{\infty} e^{(ik - k^2 + ik^3)t} \hat{f}(k) e^{-ikx} dk.$$

$$1. a) \quad \frac{\partial u}{\partial t} - 6u \cdot \frac{\partial u}{\partial x} + \frac{\partial^3 u}{\partial x^3} = 0$$

----- KdV

$$u, u_x, u_{xx} \longrightarrow 0; \quad x \longrightarrow \infty$$

$$\begin{array}{cc} \downarrow & \downarrow \\ \frac{\partial u}{\partial x} & \frac{\partial^2 u}{\partial x^2} \end{array}$$

$$\text{Show that } M(t) = \int_{-\infty}^{\infty} u(x,t) dx = \text{constant}$$

$$\text{Basically, } \frac{dM}{dt} = 0$$

$$M(t) = \int_{-\infty}^{\infty} u(x,t) dx \quad \text{--- (1)}$$

$$\Rightarrow \frac{dM}{dt} = \frac{d}{dt} \int_{-\infty}^{\infty} u(x,t) dx = \int_{-\infty}^{\infty} \frac{\partial u}{\partial t} dx \quad \text{--- (2)}$$

Make $\frac{\partial u}{\partial t}$ the subject in KdV eqn.

$$\frac{\partial u}{\partial t} = 6u \cdot \frac{\partial u}{\partial x} - \frac{\partial^3 u}{\partial x^3}$$

Substitute $\frac{\partial u}{\partial t}$ into (2)

$$\frac{dM}{dt} = \int_{-\infty}^{\infty} \left(6u \frac{\partial u}{\partial x} - \frac{\partial^3 u}{\partial x^3} \right) dx$$

$$\frac{dm}{dt} = \underbrace{6 \int_{-\infty}^{\infty} u \frac{\partial u}{\partial x} dx}_{\text{Term 1}} - \underbrace{\int_{-\infty}^{\infty} \frac{\partial^3 u}{\partial x^3} dx}_{\text{Term 2}}$$

Taking Term 1,

$$\Rightarrow \int_{-\infty}^{\infty} u \frac{\partial u}{\partial x} dx$$

We know $\frac{d}{dx} \left(\frac{1}{2} u^2 \right) = u \frac{\partial u}{\partial x}$

$$\Rightarrow \frac{1}{2} \int_{-\infty}^{\infty} \frac{d}{dx} (u^2) dx$$

$$\Rightarrow \frac{1}{2} \left[u^2 \right]_{-\infty}^{\infty}$$

Also $u \rightarrow 0$, as $x \rightarrow \infty$

$$\therefore \text{Term 1} = 6 \int_{-\infty}^{\infty} u \frac{\partial u}{\partial x} dx = 0$$

Taking Term 2,

$$\Rightarrow \int_{-\infty}^{\infty} \frac{\partial^3 u}{\partial x^3} dx = \left[\frac{\partial^2 u}{\partial x^2} \right]_{-\infty}^{\infty} = \left[u_{xx} \right]_{-\infty}^{\infty}$$

Also $u_{xx} \rightarrow 0$, as $x \rightarrow \infty$

$$\therefore \text{Term 2} = \int_{-\infty}^{\infty} \frac{\partial^3 u}{\partial x^3} dx = 0$$

$$\therefore \frac{dm}{dt} = 0, \quad m(t) = \int_{-\infty}^{\infty} u(x,t) dx \text{ is constant.}$$

1.b.

$$\hat{u}(k, t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} u(x, t) e^{ikx} dx$$

$$u(x, t) = \int_{-\infty}^{\infty} \hat{u}(k, t) e^{-ikx} dk$$

$$\text{PDE} \longrightarrow \overset{(1)}{\frac{\partial u}{\partial t}} + \overset{(2)}{\frac{\partial u}{\partial x}} = \overset{(3)}{\frac{\partial^2 u}{\partial x^2}} + \overset{(4)}{\frac{\partial^3 u}{\partial x^3}} \quad x \in \mathbb{R}, t > 0$$

$$(1) \mathcal{F}\left[\frac{\partial u}{\partial t}\right] = \frac{\partial \hat{u}}{\partial t}$$

$$(2) \mathcal{F}\left[\frac{\partial u}{\partial x}\right] = -ik\hat{u}$$

$$(3) \mathcal{F}\left[\frac{\partial^2 u}{\partial x^2}\right] = (-ik)^2 \hat{u} = -k^2 \hat{u}$$

$$(4) \mathcal{F}\left[\frac{\partial^3 u}{\partial x^3}\right] = (-ik)^3 \hat{u} = ik^3 \hat{u}$$

$$\Rightarrow \frac{\partial \hat{u}}{\partial t} - ik\hat{u} = -k^2 \hat{u} + ik^3 \hat{u}$$

$$\Rightarrow \frac{\partial \hat{u}}{\partial t} = ik^3 \hat{u} - k^2 \hat{u} + ik\hat{u}$$

$$\Rightarrow \frac{\partial \hat{u}}{\partial t} = \hat{u} (ik^3 - k^2 + ik) \longrightarrow \text{ODE}$$

$$\hat{u}(k, t) = \hat{u}(k, 0) e^{(ik^3 - k^2 + ik)t}$$

$$\hat{u}(k, 0) = \mathcal{F}[f(x)] = f(k)$$

$$\hat{u}(k, t) = f(k) e^{(ik^3 - k^2 + ik)t}$$

----- (5) soln

$$u(x,t) = \int_{-\infty}^{\infty} \hat{u}(k,t) e^{-ikx} dk$$

----- ⑥

Insert ⑤ into ⑥

$$u(x,t) = \int_{-\infty}^{\infty} f(k) e^{(ik^3 - k^2 + ik)t} e^{-ikx} dk$$

2. (based on week 2) [20 pts] The 1D wave equation is

$$\frac{\partial^2 u}{\partial t^2} = \frac{\partial^2 u}{\partial x^2}, \quad x \in \mathbb{R}, \quad t \geq 0.$$

- (a) Derive the dispersion relation $\omega = \omega(k)$ for the wave equation, compute the phase speed c_p and the group speed c_g , state whether the PDE is dispersive and/or dissipative, and justify your answer briefly.
- (b) Using Taylor series expansion about x_j , derive a five-point central finite difference approximation of $\frac{\partial^2 u}{\partial x^2}$ using the five-point stencil $\{x_{j-2}, x_{j-1}, x_j, x_{j+1}, x_{j+2}\}$ and report the lead error term.
- (c) Derive the dispersion relation of the wave equation discretized in space with three-point stencil and provide a plot by comparing the exact dispersion relation with the numerical dispersion relation.

Soln

$$\frac{\partial^2 u}{\partial t^2} = \frac{\partial^2 u}{\partial x^2}; \quad x \in \mathbb{R}, \quad t \geq 0$$

2.a) Assume plane wave solution,
 $u(x, t) = A e^{i(kx - \omega t)}$

Substituting into wave eqn

$$\Rightarrow \frac{\partial^2 u}{\partial t^2} = -\omega^2 u \quad \text{--- ①}$$

$$\Rightarrow \frac{\partial^2 u}{\partial x^2} = -k^2 u \quad \text{--- ②}$$

$$\text{①} = \text{②}$$

$$\Rightarrow -\omega^2 u = -k^2 u$$

$$\Rightarrow \omega^2 = k^2 \quad \left. \vphantom{\omega^2 = k^2} \right\} \text{Dispersion relation}$$

$$\Rightarrow \omega = \pm k$$

$$\text{Phase speed, } C_p = -\frac{\omega(k)}{k} = \frac{\pm k}{k} = \pm 1$$

$$\text{Group speed, } C_g = -\frac{\partial \omega}{\partial k} = -\frac{\partial (\pm k)}{\partial k} = \pm 1$$

∴ The PDE is ① non-dispersive (because $c_p = c_g = \text{constant}$)
 ② non-dissipative (because ω is real)

2. b) $\{x_{j-2}, x_{j-1}, x_j, x_{j+1}, x_{j+2}\}$ $\frac{\partial^2 u}{\partial x^2}$

Taylor series expansion about x_j (central)

$$\Rightarrow u_{j+1} = u(x_j + \Delta x) = u_j + \Delta x \left. \frac{\partial u}{\partial x} \right|_j + \frac{\Delta x^2}{2!} \left. \frac{\partial^2 u}{\partial x^2} \right|_j + \dots$$

$$\Rightarrow u_{j-1} = u(x_j - \Delta x) = u_j - \Delta x \left. \frac{\partial u}{\partial x} \right|_j + \frac{\Delta x^2}{2!} \left. \frac{\partial^2 u}{\partial x^2} \right|_j - \dots$$

$$\Rightarrow u_{j+2} = u(x_j + 2\Delta x) = u_j + 2\Delta x \left. \frac{\partial u}{\partial x} \right|_j + \frac{(2\Delta x)^2}{2!} \left. \frac{\partial^2 u}{\partial x^2} \right|_j + \dots$$

$$\Rightarrow u_{j-2} = u(x_j - 2\Delta x) = u_j - 2\Delta x \left. \frac{\partial u}{\partial x} \right|_j + \frac{(2\Delta x)^2}{2!} \left. \frac{\partial^2 u}{\partial x^2} \right|_j - \dots$$

Symmetrical stencil

$$\left. \frac{\partial u}{\partial x} \right|_j = a u_{j-2} + b u_{j-1} + c u_j + d u_{j+1} + e u_{j+2}$$

$$= a \left[u_j - 2\Delta x \left. \frac{\partial u}{\partial x} \right|_j + \frac{4\Delta x^2}{2!} \left. \frac{\partial^2 u}{\partial x^2} \right|_j + \dots \right]$$

$$+ b \left[u_j - \Delta x \left. \frac{\partial u}{\partial x} \right|_j + \dots \right]$$

$$+ c u_j$$

$$+ d \left[u_j + \Delta x \left. \frac{\partial u}{\partial x} \right|_j + \dots \right]$$

$$+ e \left[u_j + 2\Delta x \left. \frac{\partial u}{\partial x} \right|_j + \dots \right]$$

$$\text{Now, } a + b + c + d + e = 0 \quad (1)$$

$$(-2a - b + d + 2e) \Delta x = 1 \quad (2)$$

$$(4a + b + d + 4e) \frac{\Delta x^2}{2!} = 0 \quad (3)$$

$$(-8a - b + d + 8e) \frac{\Delta x^3}{3!} = 0 \quad (4)$$

$$(16a + b + d + 16e) \frac{\Delta x^4}{4!} = 0 \quad (5)$$

$$\begin{aligned} (3) + (4) &\Rightarrow -4a + 2d + 12e = 0 \\ &\quad -2a + d + 6e = 0 \end{aligned} \quad (6)$$

$$\begin{aligned} (3) - (4) &\Rightarrow 12a + 2b - 4e = 0 \\ &\quad 6a + b - 2e = 0 \\ &\quad b = 2e - 6a \end{aligned}$$

$$\begin{aligned} (5) \quad 16a + 2e - 6a - 6e + 2a + 16e &= 0 \\ \Rightarrow 12a + 12e &= 0 \\ \Rightarrow a &= -e \end{aligned}$$

$$\begin{aligned} \text{then } b &= 2e - 6a = -8a \\ d &= -6e + 2a = 8a \end{aligned}$$

$$\text{Finally, } -2a + 8a + 8a - 2a = \frac{1}{\Delta x}$$

$$12a = \frac{1}{\Delta x}$$

$$a = \frac{1}{12 \Delta x}$$

$$\left. \frac{\partial u}{\partial x} \right|_j = \frac{u_{j-2} - 8u_{j-1} + 0 + 8u_{j+1} - u_{j+2}}{12 \Delta x}$$

Moving on to the second derivative,

$$\frac{\partial^2 u}{\partial x^2} \Big|_j \text{ using Taylor series}$$

$$\frac{\partial^2 u}{\partial x^2} \Big|_j = a u_{j-2} + b u_{j-1} + c u_j + d u_{j+1} + e u_{j+2}$$

$$a + b + c + d + e = 0$$

$$\begin{aligned} \frac{\partial^2 u}{\partial x^2} \Big|_j &= a \left[u_j - 2\Delta x \frac{\partial u}{\partial x} \Big|_j + \frac{(2\Delta x)^2}{2!} \frac{\partial^2 u}{\partial x^2} \Big|_j - \frac{(2\Delta x)^3}{3!} \frac{\partial^3 u}{\partial x^3} \Big|_j + \dots \right] \\ &+ b \left[u_j - \Delta x \frac{\partial u}{\partial x} \Big|_j + \frac{(\Delta x)^2}{2!} \frac{\partial^2 u}{\partial x^2} \Big|_j - \dots \right] \\ &+ c u_j \\ &+ d \left[u_j + \Delta x \frac{\partial u}{\partial x} \Big|_j + \frac{(\Delta x)^2}{2!} \frac{\partial^2 u}{\partial x^2} \Big|_j + \dots \right] \\ &+ e \left[u_j + 2\Delta x \frac{\partial u}{\partial x} \Big|_j + \frac{(2\Delta x)^2}{2!} \frac{\partial^2 u}{\partial x^2} \Big|_j + \dots \right] \end{aligned}$$

Taking the coefficients,

$$a + b + c + d + e = 0$$

—(1)

$$-2a - b + d + 2e = 0$$

—(2)

$$4a + b + 2d + 4e = \frac{2}{\Delta x}$$

—(3)

$$-8a - b + d + 8e = 0$$

—(4)

$$16a + b + 2d + 16e = 0$$

—(5)

$$\textcircled{2} - \textcircled{4} \Rightarrow 6a - 6e = 0$$

$$3a - 3e = 0$$

$$a = e$$

— $\textcircled{6}$

$$\textcircled{6} \text{ into } \textcircled{2} \Rightarrow -\cancel{2e} - b + d + \cancel{2e} = 0$$

$$b = d$$

— $\textcircled{7}$

$$\textcircled{6} \& \textcircled{7} \text{ into } \textcircled{5} \Rightarrow 16a + b + 2d + 16e = 0$$

$$\Rightarrow 16e + d + 2d + 16e = 0$$

$$32e + 3d = 0$$

$$d = -\frac{32}{3}e$$

$$\therefore b = -\frac{32}{3}e$$

$\textcircled{8}$

$$\textcircled{6}, \textcircled{7}, \textcircled{8} \text{ into } \textcircled{1}$$

$$a + b + c + d + e = 0$$

$$\Rightarrow e + \left(-\frac{32}{3}e\right) + c + \left(-\frac{32}{3}e\right) + e = 0$$

$$\Rightarrow 2e - \frac{64}{3}e + c = 0$$

$$\Rightarrow c = \frac{64e}{3} - 2e = \frac{58e}{3}$$

$$\Rightarrow 4a + b + 2d + 4e = \cancel{2}\Delta x^2$$

$$\Rightarrow 4e + \left(-\frac{32}{3}e\right) + 2\left(-\frac{32}{3}e\right) + 4e = \frac{2}{\Delta x^2}$$

$$\Rightarrow 8e - \frac{96e}{3} = 8e - 32e = -24e = \frac{2}{h^2}$$

$$\Rightarrow e = -\frac{1}{12\Delta x^2}$$

3. [10pts] Consider the function

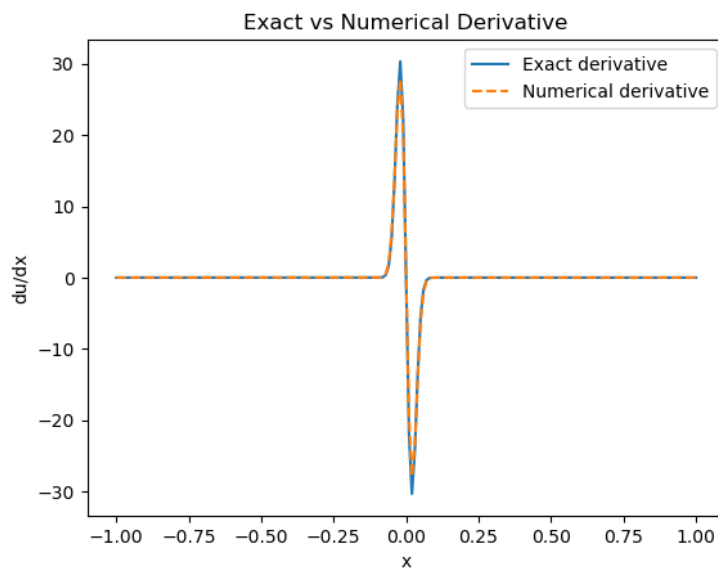
$$u(x) = e^{-\frac{x^2}{2\sigma^2}}, \quad x \in [-1, 1], \quad \sigma = 0.02.$$

- (a) Present a plot comparing exact $\frac{\partial u}{\partial x}$ with its numerical approximation.
 (b) How many grid points are needed for 2-norm error to be less than 10^{-2} ?

$$u(x) = e^{-\frac{x^2}{2\sigma^2}} \quad x \in [-1, 1], \quad \sigma = 0.02$$

$$\frac{\partial u}{\partial x} = \frac{d}{dx} e^{-\frac{x^2}{2\sigma^2}} = -\frac{x}{\sigma} e^{-\frac{x^2}{2\sigma^2}} \quad \leftarrow \text{Exact}$$

Plot



b.) $\sigma = 0.02$

$$k_{\max} \sim \frac{2\pi}{\sigma}$$

$$N = \frac{2\pi}{k\Delta x}$$

Choosing $N = 10$,

$$\Rightarrow 10 = \frac{2\pi}{k\Delta x}$$

$$\Rightarrow \Delta x = \frac{2\pi}{k \cdot 10} = \frac{2\pi}{k_{\max} \cdot 10}$$

$$\Rightarrow \Delta x = \frac{2\pi}{10 \cdot \frac{2\pi}{5}} = \frac{\pi}{\frac{10\pi}{5}} = \frac{5}{10}$$

$$\Delta x = 0.002$$

Domain, $x \in [-1, 1]$

Domain Length, $L = 1 - (-1) = 2$

$$N_{\text{grid points}} \approx \frac{L}{\Delta x} = \frac{2}{0.002} = 1000$$